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(54) **COMPOSITE DIAMOND COATED GOLF CLUBS**

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ABSTRACT

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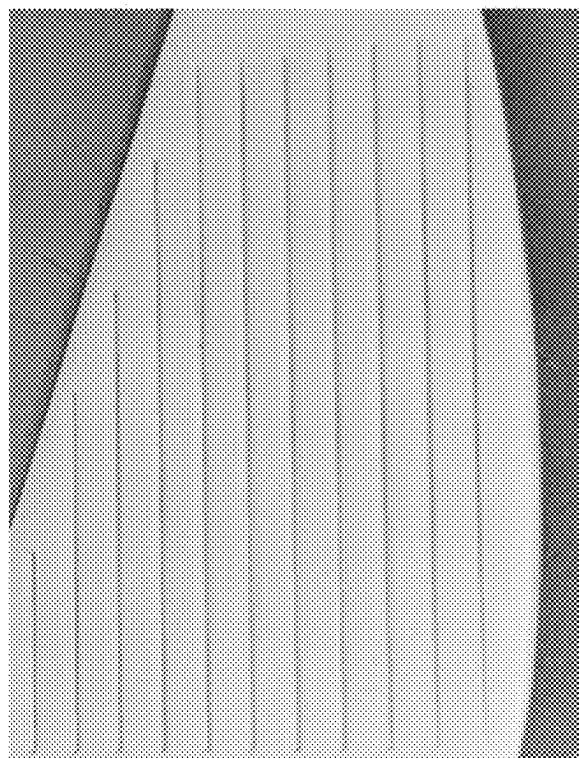
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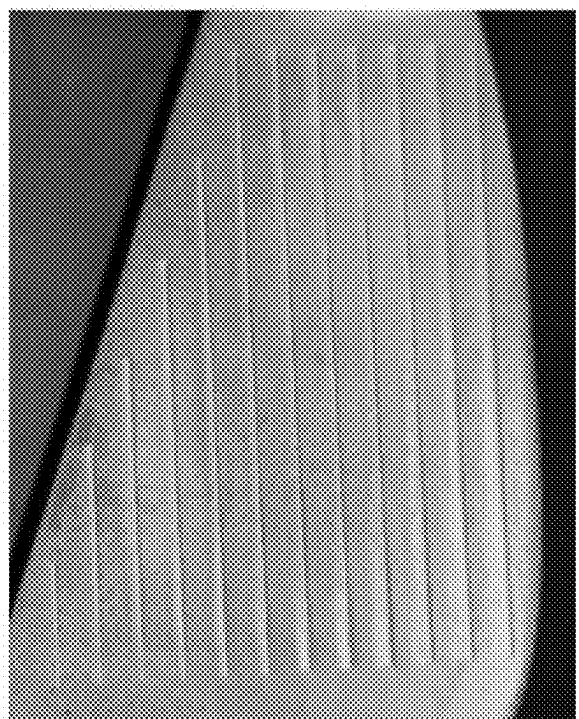
The disclosure is directed to golf clubs with composite coatings for improved performance; features; economy; environmental, health and safety; and other factors. A golf club includes at least a head and a shaft, the shaft being attached to the head, the head having a ball striking surface, where at least a central portion of the ball striking surface is covered with a nickel-based coating, said nickel-based coating further comprises diamond particulates with a uniform distribution.



FIG. 1



(R)



(L)

FIG. 2

COMPOSITE DIAMOND COATED GOLF CLUBS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority to U.S. Provisional Patent Application No. 63/630,765, filed Feb. 28, 2024, and to U.S. Provisional Patent Application No. 63/731,231, filed Apr. 15, 2024, both of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to golf club technologies, and in particular to composite diamond coated golf clubs.

BACKGROUND

[0003] Numerous varieties of plating technologies are known in the art. These technologies include electrolytic plating which is also known as electro-plating and by other terms, and electroless plating also known as chemical, autocatalytic and by other terms.

[0004] Electroless plating is a well known and established commercial/industrial process for metal plating. The metal portion of the metal salt may be selected from suitable metals capable of being deposited through electroless plating. Such metals include, without limitation, nickel, cobalt, copper, gold, palladium, iron, other transition metals, and mixtures thereof, and any of the metals deposited by the autocatalytic process described in Pearlstein, F., "Modern Electroplating", Chapter 31, 3rd Ed., John Wiley & Sons, Inc. (1974), which is incorporated herein by reference. Generally, the electroless metal in the deposited coating is a metal, a metal alloy, a combination of metals, or a combination of metals and non-metals. Such coatings are often in the form of a metal, a metal and phosphorous, or a metal and boron. The metal or metal alloy is derived from the metal salt or metal salts used in the bath. Examples of the metal or metal alloy are nickel, nickel-phosphorous alloys, nickel-boron alloys, cobalt, cobalt-phosphorous alloys, and copper alloys. Other materials such as lead, cadmium, bismuth, antimony, thallium, copper, tin, and others can be deposited to form the bath and included in the coating.

[0005] U.S. Pat. No. 10,006,126 provides a comprehensive explanation of many aspects of electroless nickel plating, and is included herein by reference.

[0006] The salt component of the metal salt may be any salt compound that aids and allows the dissolution of the metal portion in the bath solution. Such salts may include without limitation, sulfates, chlorides, acetates, phosphates, carbonates, and sulfamates, among others.

[0007] The reducing agents are electron donors. When reacted with the free floating metal ions in the bath solution, the electroless reducing agents reduce the metal ions, which are electron acceptors, to metal for deposition onto the article. The use of a reducing agent avoids the need to employ a current, as required in conventional electroplating. Common reducing agents are sodium hypophosphite, sodium borohydride, n-dimethylamine borane (DMAB), n-diethylamine borane (DEAB), formaldehyde, and hydrazine.

[0008] Certain materials may be used in electroless plating baths where these materials serve two or more roles in the

plating bath. For example, instead of using the typical combination of nickel sulfate as a metal salt and sodium hypophosphite as a reducing agent, it is possible to use nickel-hypophosphite in an electroless nickel plating bath. Nickel-hypophosphite, however, is very expensive and not widely used commercially due to its impractical cost.

[0009] Electroless nickel (EN) is one of the most commercialized varieties of electroless plating. It is an alloy of nominally 86-99% nickel and the balance with phosphorous, boron, or a few other possible elements. Electroless nickel is commonly produced in one of four alloy ranges: low (1-5% P), medium (6-9% P), or high (10-14% P) phosphorous, and electroless nickel-boron with 0.5-5% B. Each variety of electroless nickel thus provides properties with varying degrees of hardness, corrosion resistance, magnetism, solderability, brightness, internal stress, lubricity, and other properties. All varieties of electroless nickel can be applied to numerous articles, including metals, alloys, and nonconductors.

[0010] Electroless composite technology is a more recent development as compared to electrolytic composite technology. The fundamentals of composite electroless plating are documented in a text entitled "Electroless Plating Fundamentals and Applications," edited by G. Mallory and J. B. Hajdu, Chapter 11, published by American Electroplaters and Surface Finishers Society (1990).

[0011] The plating of articles with a composite coating bearing finely dispersed divided particulate matter is well documented. The inclusion of finely divided particulate matter within metallic matrices can significantly alter the properties of the coating with respect to properties such as wear resistance, lubricity, friction, thermal transfer, and appearance.

[0012] The co-deposition of particles in composite electroless plating can dramatically enhance existing characteristics and even add entirely new properties. These capabilities have made composite electroless coatings advantageous for a variety of reasons including, but not limited to, increased utility in conditions requiring less wear, lower friction, lubrication, indication, authentication, thermal transfer, insulation, higher friction, and others. Composite electroless coatings with nickel provide an additional environmental advantage over conventional electroless nickel coatings, which do not include particulate matter, in that the particles within composite electroless nickel coatings reduce the amount of nickel alloy used. Such nickel based composite coatings are also an alternative to chromium based coatings which pose certain health and environmental challenges.

[0013] Particulate matter suitable for practical composite electroless plating may be from nanometers up to approximately 100 microns in size. The specific preferred size range depends on the application involved.

[0014] The particulate matter may be selected from a wide variety of distinct matter, such as but not limited to ceramics, glass, talcum, plastics, diamond (polycrystalline or monocrystalline types, natural or manmade by a variety of processes), graphite, oxides, silicides, carbonate, carbides, sulfides, phosphate, boride, silicates, oxyates, nitrides, fluorides of various metals, as well as metal or alloys of boron, tantalum, stainless steel, molybdenum, vanadium, zirconium, titanium, tungsten, as well as polytetrafluoroethylene (PTFE), silicon carbide, boron nitride (BN), aluminum oxide, graphite fluoride, tungsten carbide, talc, molybdenum

disulfide (MOS), boron carbide and graphite. The boron nitride (BN), without limitation, may be hexagonal or cubic in orientation.

[0015] For increased friction on the surface of a resultant coating and/or increased wear resistance, hard particulates, such as but not limited to diamond, carbides, oxides, and ceramics, may be included in the plating bath. Application of an overcoat of a conventional plated layer on top of the composite plated layer is also done in the field in order to further embed the particulate matter within the coating.

[0016] For increased lubrication or reduction in friction in the resultant coating, "lubricating particles," such as polytetrafluoroethylene (PTFE), boron nitride (BN), talc, molybdenum disulfide (MOS), graphite or graphite fluoride among others may be included in the plating bath. These lubricating particles may embody a low coefficient of friction, dry lubrication, improved release properties, and/or repellency of contaminants such as water and oil.

[0017] For light emitting properties in the resultant coating, particulates with phosphorescent properties such as, but not limited to, calcium tungstate may be included in the plating bath.

[0018] For identification, authentication, and tracking properties in the resultant coating, various particulate and solid materials may be included in the plating bath so they will be incorporated into the coating and detectable either visually, under magnified viewing, or detection with a suitable detector. The need for such identification, authentication, and tracking properties are useful to golf equipment which can be susceptible to counterfeiting and other forms of product and intellectual property issues. Therefore, the present disclosure includes the potential to meet this unmet need.

[0019] The inclusion of insoluble particulate matter in composite electroless baths introduces additional instability. To overcome the extra instability due to the addition of insoluble particulate matter to the bath, such as described in U.S. Pat. No. 6,306,466, the general use of particulate matter stabilizers (PMSs) is believed to isolate the finely divided particulate matter, thereby maintaining the particular matter's "inertness". Such PMSs are well-known, and include, without limitation, sodium salts of polymerized alkyl naphthalene sulfonic acids, disodium mono ester succinate (anionic and nonionic groups), fluorinated alkyl polyoxyethylene ethanols, tallow trimethyl ammonium chloride, and any of the PMS disclosed in U.S. Pat. No. 6,306,466, which is incorporated herein by reference.

[0020] The electroless metallizing bath may also contain one or more complexers, also known as complexing agents. A complexing agent acts as a buffer for reasons which may include pH control and maintaining control over the "free" metal salt ions in the solution, all of which aids in sustaining a proper balance in the bath solution.

[0021] The electroless metallizing bath may further contain a pH adjuster to also help control pH levels in the bath. Suitable pH adjusters may buffer the plating bath at a desired pH range.

[0022] Some materials may serve one or more functions within an electroless plating bath. For example, ammonium hydroxide is both a pH adjuster as well as a complexer; cadmium, aluminum, copper and others materials are both a stabilizer and a brightener, lactic acid is both a complexer and a brightener, some sulfur compounds like thiourea are

both stabilizers and accelerators depending on concentration, and there are other multipurpose ingredients useful in electroless plating baths.

[0023] Ingredients typical in electroless plating and useful in the present disclosure include, but are not limited to the following materials in the following general categories:

Complexers

[0024] Acetic Acid, Alanine-beta, Aminoacetic Acid, Ammonium Bicarbonate, Ammonium Carbonate, Ammonium Chloride, Ammonium Hydroxide, Boric Acid, Citric Acid, Citrates, EDTA, Ethylenediamine, Fluoboric Acid, Glycerine, Glycine, Glycolic Acid, Glycolic Acid Salts, Hydroxyacetic Acid, Lactic Acid, Maleic Anhydride, Malic Acid, Malonic Acid, Orthoboric Acid, Oxalic Acid, Oxalic Acid Salts, Propionic Acid, Sodium Acetate, Sodium Glucoheptonate, Sodium Hydroxyacetate, Sodium Isethionate, Sodium or Potassium Pyrophosphate, Sodium Tetraborate, Succinic Acid, Succinate Salts, Sulfamic Acid, Tartaric Acid, Triethanolamine, Monocarboxylic Acids, Dicarboxylic Acids, Hydrocarboxylic Acids, Alkanolamines, and combinations and variations of such materials.

Stabilizers

[0025] 2 Amino-Thiazole, Antimony, Arsenic, Bismuth Compounds, Cadmium Compounds, Lead Compounds, Heavy Metal Compounds, Iodobenzoic Acid, Manganese Compounds, Mercury Compounds, Molybdenum Compounds, Potassium Iodide, Sodium Isethionate, Sodium Thiocyanate, Sulfur Compounds, Sulfur Containing Aliphatic Carbonic Acids, Acetylenic Compounds, Aromatic Sulfides, Thiophenes, Thionaphthalenes, Thioarols, Thiodipropionic Acid, Thiodisuccinic Acid, Tin Compounds, Thallium Sulfate, Thiodiglycolic Acid, Thiosalicylic Acid, Thio-urea, and combinations and variations of such materials.

Brighteners

[0026] Aluminum, Antimony Compounds, Cadmium Compounds, Copper, Lactic Acid, and combinations and variations of such materials.

pH Controllers

[0027] Ammonium Bicarbonate, Ammonium Carbonate, Ammonium Chloride, Ammonium Hydroxide, Potassium Carbonate, Potassium Hydroxide, Sodium Hydroxide, Sulfamic Acid, Sulfuric Acid, and combinations and variations of such materials.

Particulate Matter Stabilizers (Dispersants, Surfactants, Wetters)

[0028] Sodium salts of polymerized alkyl naphthalene, disodium mono ester succinate (anionic and nonionic groups), fluorinated alkyl polyoxyethylene ethanols, tallow trimethyl ammonium chloridesulfonic acids, disodium mono ester succinate (anionic and nonionic groups), fluorinated alkyl polyoxyethylene ethanols, tallow trimethyl ammonium chloride, and any of the PMS disclosed in U.S. Pat. No. 6,306,466, which is incorporated herein by reference, and combinations and variations of such materials.

Buffers

[0029] Borax, Boric Acid, Orthoboric Acid, Succinate Salts, and combinations and variations of such materials.

Reducing Agents

[0030] DMAB, DEAB, Hydrazine, Sodium Borohydride, Sodium Hypophosphite, and

[0031] combinations and variations of such materials.

Accelerators

[0032] Fluoboric Acid, Lactic Acid, Sodium Fluoride, Anions of some mono and di carboxylic acids, fluorides, borates, and combinations and variations of such materials.

Metal Salts

[0033] Cobalt Sulfate, Copper Sulfate, Nickel Sulfate, Nickel Chloride, Nickel Sulfamate, Nickel Acetate, Nickel Citrate, and combinations and variations of such materials.

[0034] Historically electroless nickel and composite electroless plating processes have included heavy and/or toxic metals in the plating bath to overcome the inherent instability of the plating bath. Lead has been the most commonly used material to serve this purpose. Cadmium has also been used widely over the years as a brightener for electroless nickel coatings. But this incorporation of heavy metals into the plating baths presents multiple challenges. The heavy metals must be added in a sufficient amount to prevent the decomposition of the plating bath, but an increased concentration beyond the necessary level required to prevent the decomposition results in cessation or reduction of the plating rate. Increasingly stringent rules and regulations that restrict or prohibit the use of heavy metals, such as the Removal of Hazardous Substances (RoHS) and End-Of-Life Vehicle (ELV) Regulations. However, U.S. Pat. Nos. 7,744,685 and 8,147,601 disclose stable composite electroless nickel plating baths without the use of heavy and/or toxic metals. These patents are included herein by reference.

[0035] The electroless nickel and composite electroless nickel coatings of the present disclosure may contain heavy metals or may be essentially free of heavy metals, which means that no such heavy metal is added to the plating bath and/or the heavy metal concentration should be no more than a level that would cause the coating on articles plated in said bath to have a heavy metal concentration in excess of any relevant regulations. The coatings of the present disclosure may also contain heavy metals less toxic and/or subject to fewer regulations than lead, cadmium and others.

[0036] In recent years, there has been a growing desire within the plating industry to avoid the use of ammonium hydroxide. Ammonium hydroxide is an effective complexing agent and pH adjuster. Ammonium hydroxide, however, is objectionable to some plating shops due to environmental, health and/or safety regulations, smell, and the difficulty it causes in the ability to remove the nickel from the plating bath at the end of the bath's life because it is such a strong complexing agent. Storage and handling of ammonium hydroxide is also problematic as it can cause storage drums and other containers to bloat, it emits a very noxious odor experienced when opening a container, pumping, and transporting ammonium hydroxide, and causes a strong reaction when added to a hot plating bath unless the extra step of diluting the ammonium hydroxide by 50 percent by volume

or more is performed in advance. Specially designed respirators are needed when handling ammonium hydroxide. It is therefore desirable to have a solution for an electroless nickel plating bath where this solution is free of ammonium hydroxide, and whereby the user or plater has the ability to use a material other than ammonium hydroxide as an auxiliary solution to maintain the pH of the plating bath during usage. The present disclosure includes the possibility of producing the composite coatings with or without the use of ammonium hydroxide or other ammonia compounds.

[0037] In recent years, there has been a growing desire within the plating industry to use lower concentrations of metal salts in the plating baths. The primary justifications for this alternative to the conventional concentrations of metal salts in the plating baths are to 1) reduce the drag out of the metal salts from the plating baths to the subsequent rinse tanks and thereby reduce the amount of metal salts that need to be captured in subsequent waste treatment of the rinse water facilitating better environmental practices, 2) reduce the amount of metal salts that are essentially wasted when the plating bath comes to the end of its useful life and the bath is waste treated or otherwise disposed of, and 3) improve the quality of the plating by lowering the amount of metal salts in the bath which have the potential to precipitate or react in the bath in ways other than the desired reduction and deposition onto articles immersed in the plating bath for the purpose of plating, especially effective in reducing shelf roughness, 4) lowering the cost to make up a plating bath, 5) extend plating bath life, especially when plating onto aluminum substrates, 6) increase reducing agent efficiency, and 7) contain less metal and other substances in the mist emanating from the plating bath. An example of this practice is in the electroless nickel plating field where some platers are using plating baths with less than the traditional 6 grams per liter of nickel metal in the bath, for example, 3 grams per liter. The background and justification for using electroless nickel plating baths with a reduced nickel content is well documented in: <http://www.pfonline.com/articles/fifth-generation-reduced-ion-electroless-nickel-systems>.

When applied to electroless nickel plating systems, the present disclosure can include composite coatings manufactured with a traditional concentration of 6 grams per liter of nickel metal in the plating bath, 3 grams per liter of nickel metal in the plating bath, or other concentrations.

[0038] In addition, in recent years, health and environmental concerns have been raised about the inclusion of certain materials such as perfluorooctane sulfonate (PFOS) and perfluorooctanoic acid (PFOA) that may be used in plating systems including composite plating systems, including those with PTFE. PFOS may be contained in certain particulate matter stabilizers (PMSs) useful for electroless plating. The present disclosure therefore includes coatings made from compositions, baths, and methods for plating that may contain PFOA and/or PFOS, or may be free, or have only trace amounts of PFOA and/or PFOS.

[0039] While many elements of the EN plating chemistry, process, and industry have evolved, one essential aspect of the technology has remained relatively unchanged since the early style baths were surpassed by formulations that were easier and more reliable to operate. This aspect is the method to make up and maintain the EN plating bath. Make up of an EN bath involves combining the ingredients required to create a bath that is ready to be used for its intended purpose. Maintenance or replenishment of the EN bath involves

replacing the chemical elements of the bath that have been depleted from the bath as plating occurs from the bath onto articles immersed in the bath.

[0040] While it is possible to make up and replenish a plating bath by adding the desired amount of each individual ingredient to form a solution, the established method to make up and replenish a plating bath is to combine three or more separate pre-made solutions with water.

[0041] When three solutions are used, it is common in the field to make up an EN bath with an "A" solution and a "B" solution and water. The A solution typically contains the metal salt (for example, nickel sulfate), may contain other ingredients, and accounts for five to six percent of the volume of the plating bath. The B solution typically contains the reducing agent (for example, sodium hypophosphite), other functional ingredients like stabilizers, brighteners, pH buffers, chelators, complexing agents, accelerators, particulate matter stabilizers, etc., and accounts for fifteen to twenty percent of the volume of the plating bath. The balance, typically about eighty percent of the volume of the plating bath, is made up of water plus the possibility of an acid or base to adjust the pH of the EN bath before it is heated to the desired temperature and used for plating. The water is typically deionized water. That is, the initial bath is comprised of the A solution, the B solution, water, and potentially a pH adjuster, where the pH adjuster may be introduced into the water before being combined with A and B.

[0042] The use of multiple plating compositions as described herein, is referred to as a "plating bath system".

[0043] As the bath is used, it needs to be replenished. The EN bath is then typically replenished with the A solution as well as a "C" solution. The C solution is typically similar to the B solution, containing the reducing agent (for example, sodium hypophosphite), other functional ingredients like stabilizers, brighteners, pH buffers, chelators, complexing agents, accelerators, particulate matter stabilizers, etc., but the specific combination and concentration of these materials are in different concentrations in the C solution than they are in the B solution. The reason for the difference of concentrations of these materials is the difference in the consumption or depletion rate of each material from the initial make up concentration due to the plating reaction. The C solutions are typically formulated to be used in a convenient ratio to the A solutions, for example one part A solution plus two parts C solution; or for example one part A solution plus one part C solution.

[0044] When more than two solutions are used, such as the Addplate™ concentrate systems sold by Surface Technology, Inc. of Ewing, NJ, it is common in the field to make up an EN bath with three solutions such as 1) an "M" solution, 2) a solution of nickel sulfate, and 3) a solution of sodium hypophosphite, plus water. The M solution typically contains the functional ingredients like stabilizers, brighteners, pH buffers, chelators, complexing agents, accelerators, particulate matter stabilizers, etc., and accounts for eight to ten percent of the volume of the plating bath. The nickel sulfate and sodium hypophosphite solutions typically account for four and a half percent each of the volume of the plating bath. The balance, typically about eighty-two percent of the volume of the plating bath, is made up of water plus the possibility of an acid or base to adjust the pH of the EN bath before it is heated to the desired temperature and used for plating. The water is typically deionized water. The EN bath is then typically replenished with an "R" solution as well as

the nickel sulfate and sodium hypophosphite solutions. The R solution is typically similar to the M solution, containing the functional ingredients like stabilizers, brighteners, pH buffers, chelators, complexing agents, accelerators, particulate matter stabilizers, etc., but the specific combination and concentrations of these materials are in different concentrations in the R solution than they are in the M solution. The reason for the difference of concentrations of these materials is due to the difference in the consumption or depletion rate of each material from the plating bath during usage of the plating bath and the plating reaction. The R solutions are formulated to be used in a convenient ratio to the nickel sulfate and sodium hypophosphite solutions, for example one part nickel sulfate solution plus one part sodium hypophosphite solution plus one part R solution; or for example one part nickel sulfate solution plus one part sodium hypophosphite solution plus one half or one third part R solution.

[0045] Some companies in the plating industry have offered and/or used systems where the bath can be made up of one single component instead of two, three or more. But in none of these systems is it possible to replenish that same bath with the same make up solution for ongoing maintenance of the bath over the bath's life while providing proper bath stability and plating quality.

[0046] It is possible, especially as would be evident to one skilled in the art from understanding the present disclosure, to produce coatings from an electroless plating bath with one component used alone to make up the plating bath and a second component used alone to replenish the plating bath.

[0047] When discussing the materials and solutions used in the make-up and replenishment of electroless plating baths, and if the system is a one, two, three, four or more solution system, it is customary in the field to count the number of solutions containing the primary functional ingredients such as metal salts, reducing agents like stabilizers, brighteners, pH buffers, chelators, complexing agents, accelerators, particulate matter stabilizers, etc., and mixtures thereof. The addition of any other ingredients to the plating bath is not considered an additional solution. For example, the addition of materials such as ammonium hydroxide, other hydroxides, carbonates and the like to adjust the pH of the plating bath are not considered a solution in the same way as a typical A, B, C, M or R solution is counted in the system. These materials are considered auxiliary solutions. Solutions of additional stabilizers, brighteners, accelerators, PMSs, and other materials may also be used as auxiliary solutions to modify the plating bath for specific purposes, often for episodic purposes rather than consistent uses. If such materials were needed for consistent, routine purposes in the plating bath, they might be incorporated into one or more of the primary solutions such as the A, B, C, M or R solutions. Similarly, the addition of particulate matter, in powder, liquid dispersion, or other form, is also considered an auxiliary material or solution, and is not considered a solution or component in the same way as a typical A, B, C, M or R solution is considered as a solution in the system.

[0048] Consequently, it would be beneficial for a single solution usable for both initial and replenishment purposes to produce coatings in the present disclosure.

[0049] The typical operation of an electroless plating bath consists of the following steps. First, a plating bath is made up traditionally as already discussed in this disclosure. The plating bath is then heated by any of a number of mecha-

nisms to reach a desired operating temperature. Articles for plating are then cleaned and otherwise pretreated according to their base metal(s) and condition, and immersed into the plating bath. While the articles are being plated for a time commensurate with the plating rate of the plating bath and the desired thickness of the plating onto the articles, the temperature and pH of the plating bath are typically monitored and maintained at desired levels. During or after the plating of the articles, the plating bath is analyzed to determine the concentration of certain components in the plating bath. Typically this analysis includes the concentration of the metal of the metal salt in the plating bath, and this is accomplished by wet chemistry or by instrumental analysis. Based on the concentration of this metal in the plating bath, the plating bath is traditionally replenished with two or more solutions containing the ingredients needed to replace what has been depleted onto the articles. This replenishment can be added to the plating bath by pouring, pumping, or other means. Analysis of other components such as reducing agents and stabilizers in the plating bath can be accomplished, but is much less common, and therefore increases the potential for the ratio of ingredients to become imbalanced with the metal salt and other ingredients in the plating bath.

[0050] The present disclosure is directed to the use of such plating processes, solutions, and coatings uniquely applied to golf clubs.

BRIEF DESCRIPTION OF THE DRAWINGS

[0051] The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

[0052] The disclosure will be understood more fully from the detailed description given below and from the accompanying drawings of various embodiments of the disclosure. The drawings, however, should not be taken to limit the disclosure to the specific embodiments, but are for explanation and understanding only.

[0053] FIG. 1 illustrates a golf club head according to an embodiment of the disclosure.

[0054] FIG. 2 illustrates a golf club head under ultraviolet light (left) and conventional LED ambient light (right) according to another embodiment of the disclosure.

SUMMARY OF THE PRESENT DISCLOSURE

[0055] The present disclosure is directed to golf equipment with improved performance due to composite coatings. The present disclosure includes the coatings, the bath used for the coatings, and the coated golf clubs. The present disclosure includes the use of any of a number of types of diamond for coating, where diamond particulate matter is included in a bath, particulate matter stabilizers are also included to aid in suspending the diamond in the bath thereby facilitating a more uniform coating on the golf club. The bath further may include different levels of phosphorous, where the level of phosphorous can be classified as low, high or medium, each of which may bring different properties to the coating. The contents of the bath of the present disclosure can be absent PFOS, PFOA, other PFAS materials and heavy metals, such as but not limited to lead and cadmium. Also, the coatings of the present disclosure

may be applied to any or all of the golf club shaft, head, or hosel. In addition, different clubs may have different coatings applied, based at least in part on the purpose of the club. However, regardless, the present coatings provide significant improvement over the prior art in durability and ball performance, such as but not limited to accuracy and distance.

DETAILED DESCRIPTION OF THE PRESENT DISCLOSURE

[0056] Golf equipment, especially golf clubs, are designed for optimal performance. There is, in fact, a market of avid consumers interested in using golf clubs and other golf products for optimal performance. In U.S. Pat. No. 5,029, 865 ("Kim"), Kim describes some of the performance properties desired on golf clubs, absent many details. This Kim patent is included herein by reference. Kim discloses a particular high phosphorous coating, further containing one or more of lead and cadmium and Kim is silent as to including particulate matter stabilizers. Kim also discloses use on plates, not golf clubs, yet asserts, without proof, that the coatings could be used on golf clubs to improve performance. Kim, however, fails to demonstrate any performance benefits on actual golf clubs. Further, Kim did not disclose many golf club performance factors that are desirable for golf clubs. A more thorough list of performance features desirable on golf clubs includes, but is not limited to:

[0057] Longer ball distance.

[0058] Greater directional accuracy of the golf ball shot

[0059] Greater consistency of the golf ball shot.

[0060] Optimal spin of the golf ball by the golf club for iron and wood shots.

[0061] Optimal spin of the golf ball by the golf club for wedge shots.

[0062] Wear resistance and durability of the golf club profile, including but not limited to on irons, woods, wedges, and putters.

[0063] Heat transfer of the energy of the golf shot from a golf club.

[0064] Time of impact between the golf club and the golf ball

[0065] Corrosion resistance of the golf club.

[0066] Longer life of the golf club.

[0067] Conformity to rules established by agencies such as the United States Golf Association (USGA), Players Golf Association (PGA), and others that govern rules on equipment and other parameters of the sport.

[0068] Environmental, health and safety considerations in the manufacture of golf clubs. Economy in the manufacture of golf clubs

[0069] Improved bonding between the golf club head and the golf club shaft.

[0070] The present disclosure is directed to golf clubs comprising composite coatings on at least one of a club shaft, club head, or hosel as described herein. The club head of each golf club has a ball striking surface which is the surface or area of a golf club that is intended to hit the golf ball and impart the desired motion of the ball. The ball striking surface is often called the face or other terms of a golf club. For this reason, the application of a coating onto the ball striking surface or face of a golf club is of particular relevance to the present disclosure. Consequential to the coatings of the present disclosure, the golf clubs of the present disclosure may have enhanced performance in one

or more ways as listed herein. Further, explanations are disclosed herein as to how the present disclosure surpasses the utility and scope of Kim and any other prior art. Whereas Kim discloses use of hardened coating on specific clubs and portions of clubs, the Kim disclosure is limited in detail, particularly with respect to the processes used for coating, the coatings themselves, and the solutions used for coating. The present disclosure overcomes these limitations by providing added novel detail as well as by providing specific improvements in performance over the Kim descriptions, at least in part by utilizing alternate coating solutions and techniques, resulting in differing coating compositions and thicknesses.

[0071] Importantly, the present disclosure is distinguishable from Kim at least in terms of the solutions used for electroless nickel coating and the content of such solutions.

[0072] Among the improvements in the present disclosure are:

[0073] Use of PMSs in the baths used for coating golf clubs. The novel disclosure of use of PMSs relative to golf clubs results in a uniform diamond distribution in the coating, avoidance of undesirable over or under roughness in the coating, avoidance of diamond particles getting plated (instead of just co-deposited), coating consistency, and process efficiency and economy. Specifically, as PMSs isolate the finely divided particulate matter in the plating bath, and maintain the particular matter's "inertness" in the plating bath, the PMSs serve to help make the resulting coatings uniform in roughness at specified desired levels of roughness. The present disclosure, more particularly the chemical composition of the plating bath and the plating bath itself, controls the coating application to achieve the desired level and uniformity of roughness on a case-by-case, or type of club, basis. Such a result would be unachievable by Kim based on the Kim disclosure, as there is no evidence of use of PMSs by Kim, nor is his starting point a golf club (his example related only to a steel plate). Without the use of PMSs in composite plating baths, undesired and uncontrolled roughness can be formed by a number of mechanisms, including the following, any or all of which may likely have happened in Kim's approach as described:

[0074] Avoidance of plating the particles themselves which would increase the size of the particles, which could then be co-deposited into the composite plated layer on the golf club.

[0075] Avoidance of plating the particles themselves which then result in added metallic surface area in the plating bath that can cause instability of the plating bath by depleting the stabilizers and other ingredients in the plating bath that are present to maintain plating bath stability.

[0076] Avoidance of plating the particles themselves which then result in metallic particles that can cause "plate out" of the tank and/or decomposition of the plating bath. Either of these effects would cause non-uniform roughness in the plated layer on the golf club, thereby adversely affecting the club's consistency, and take away from the ability of the plating bath to be used consistently and economically in a commercial application.

[0077] Avoidance of the agglomeration of the particulate matter in the plating bath, which could then cause inconsistencies in the coating's composition. This can include irregular roughness in the coating if the agglomerations of particulate matter get co-deposited in clusters. It could also cause voids in certain areas of the coating if the agglom-

erations settle on the surface of the golf club during the process, without being co-deposited, but inhibit the desired co-deposition of non-agglomerated particles from the plating bath.

[0078] Moreover, the use of PMSs in the composite plating of golf clubs enables the use of a much lower concentration of diamond particles in the plating bath than used by Kim in his simplistic trial. Kim used 28 grams of diamond per liter of the plating bath in his trial. That is an exceedingly high amount of diamond for such a plating bath, and such a high amount of diamond, especially without the use of PMSs, increases the vulnerability of instability (and the negative effects noted above including particle agglomeration and coating roughness) of the plating bath, uncontrolled roughness, and higher cost.

[0079] It is well known in the field of commercial plating that the use of PMSs further aid in the production of plated layers that are smoother and more consistent. The PMSs serve to mitigate pitting in the plated layer. Pitting is one form of roughness that is often caused by the dwelling of hydrogen bubbles on the surface of the object during plating. PMSs help to release such bubbles from the surface of the object during plating before the bubbles are able to cause such pitting, and/or to minimize the degree of such pitting. PMSs can also reduce or avoid nodularity in the plated layer. PMSs are used in many varieties of electroless nickel plating, including high phosphorous electroless nickel, which is often used for corrosion resistance where the avoidance of pits in the coating are important to avoid corrosion. Similarly, PMSs are often used in high phosphorous electroless nickel for electronics applications such as computer storage devices and optical equipment to avoid nodularity that can interfere with the intended purpose of the coating and part.

[0080] Also novel in the present disclosure relative to golf clubs is the use of different phosphorous levels in the plating bath. The level of phosphorous impacts coating hardness, wear resistance of the coating, ductility, corrosion resistance, and the plating rate, which can impact process efficiency and economy.

Longer Ball Distance

[0081] Unlike Kim, the present disclosure provides data demonstrated the increased ball distance resulting from golf balls hit by actual golf clubs in controlled experiments, where the club comprises the composite coatings of the present disclosure. Kim only tests panels with and without coating, but the difference in the dynamics between golf balls rebounding from a smooth plate due to the force of gravity is vastly different than the dynamics of hitting a golf ball with a golf club with a defined shape and hitting surface including grooves swinging at high speeds in a relatively horizontal direction. That is, in one embodiment of the present disclosure, the composite coating follows the contours of a club head's grooves. Golf balls in actual use are essentially launched into flight by a golf club. While not wanting to be bound by theory, the dimples on golf balls serve a similar lift purpose as do wings on an airplane. Bouncing balls on flat panels does not replicate the actual mechanism of a golf club hitting a golf ball. The present application includes examples demonstrating increased distance in terms of both the carry (or flight of a golf ball) and the total distance of the golf ball after rolling in an actual fairway (grass covered area of a golf course) due to treat-

ment of a golf club with a composite coating as disclosed in the present disclosure. The present disclosure also discloses the roughness of the surface of the composite coating applied to the examples herein, which is a factor not considered nor disclosed by Kim.

Greater Directional Accuracy of the Golf Ball Shot

[0082] Kim is silent on the accuracy and linearity of the golf ball from golf clubs because he never tested this mechanism. In the testing of the present disclosure, the directional accuracy of the balls hit by golf clubs with the coating of the present disclosure was tested and results documented. This testing includes the shots from both a club designed for hitting the ball long distances (a metal wood) and the shots from a wedge style club with a higher loft that is designed for shorter distance shots in the play of golf.

Greater Consistency of the Golf Ball Shot

[0083] In the examples of the present disclosure relative to consistency of distance, the present disclosure includes demonstrated greater consistency in the distance of the golf ball from a golf club with the composite coating of the present disclosure compared to an identical but uncoated golf club. This increased consistency is expressed in a lower standard deviation in both carry and total distance from a golf club treated with a composite coating of the present disclosure. The greater wear resistance of the coatings of the present disclosure further provides for a greater consistency over a longer period of time. The greater wear resistance is achieved in the present disclosure by factors including, but not limited to, higher diamond density in the coating, uniformity of the diamond in the coating, less particle size distribution of the diamond particles, heat treatment of the coating, among other factors.

[0084] Optimal spin of the golf ball by the golf club for iron and wood shots

[0085] Attention to the surface finish of the composite coatings in the present disclosure includes the design of coatings for irons and woods to have a defined roughness that will not cause undesirable spin that can affect the direction of the ball and distance of the ball.

[0086] Optimal spin of the golf ball by the golf club for wedge shots

[0087] This is demonstrated in the present disclosure as disclosed in an example herein. Kim is silent on the negative effect of spin on golf balls hit by long distance woods and irons where the spin can cause the ball to travel to the left or right instead of straight and also reduce the distance of the shot. Likewise, Kim is silent on the beneficial use of spin on wedge type golf clubs where spin can help the ball land accurately (typically on the putting green) and enable a subsequent motion of the ball to roll forwards or backwards to a targeted location.

[0088] Importantly, the present disclosure includes a variety of different concentrations of chemicals, and in some cases somewhat different chemical mixes, which could differ based on being applied to different types of golf clubs where the club is intended for different purposes. For example, the roughness of the coating might vary based on desired spin. In these cases, the percentage of diamond in the solution, the type of diamond, the size of diamond, and the concentration of particulate matter stabilizer might vary.

[0089] Wear resistance and preservation of the golf club profile on irons, woods, wedges, and putters

[0090] Kim does not address this important point at all. Wear resistance is essential not just for a longer lifetime of the golf club, but it is important as well for the performance to be consistent during the lifetime of the golf club. The very specific surface finish, profile, and geometry of the hitting face as well as other areas of a golf club are all integral to the consistent and long term performance of the golf club. The present application includes an example of the superior wear resistance of a golf club treated with a composite coating according to the present disclosure.

[0091] The electroless nickel plating bath in Kim (limited to Enthone 415) is different and inferior to the electroless nickel plating baths of the present disclosure in numerous ways that make the present disclosure able to meet unmet needs on the coating of golf clubs. The Enthone 415 plating bath is a high phosphorous electroless nickel alloy. The present disclosure includes the disclosure examples with medium phosphorous electroless nickel, which results in harder and more durable coatings than high phosphorous alloys. The Enthone 415 product also contains lead. The present disclosure can be and has been made without the heavy metal lead. This is significant for environmental, health and safety concerns, as well as improved regulatory compliance. The ability of the present disclosure to produce coatings on golf clubs without lead in the coating makes the present disclosure capable of meeting numerous regulations such as Removal of Hazardous Substances (RoHS) and the like. Kim is silent on the stability and ongoing use potential of the plating bath used in his example. In the present disclosure, the plating baths have been demonstrated to operate at a commercially viable plating rate, stability, and lifetime (measured in MTOs or Metal Turn Overs). In the present disclosure, a pre-manufactured dispersion product of diamond particles combined with Particulate Matter Stabilizers (PMSs) was used in the plating baths. This fact, different than Kim who merely added diamond to a plating bath, enables the high level of diamond particle density and uniformity of the present disclosure, which is not achieved by Kim, and which is essential for optimal performance of the golf equipment. These significant differences in the type and operation of the plating baths between the present disclosure and the primitive example of Kim are also important for making the present disclosure more cost effective and economically viable for commercial use of these systems for golf equipment.

Time of Impact Between the Golf Club and the Golf Ball

[0092] By providing a higher density of diamond or other hard particles in the composite coating on golf clubs than allowable using Kim's described coating, more energy from the swing of the golf club is transferred to the ball. This phenomenon increases the duration of contact and accelerates the release of the golf ball from the face of the golf club. Therefore more energy is transferred to the ball. By reducing the time of impact between the golf ball and the golf club, there is less friction between golf ball and the face of the golf club. Moreover, by reducing the time of impact and friction as the golf ball is hit by the golf club, there is less spin imparted onto the golf ball by the golf club, further improving performance. Less spin for distance shots such as from woods and irons will provide greater accuracy in the resulting location of the golf ball. The surface roughness of the

impact area of a golf club face is important for performance. Higher and lower roughnesses can have a significant effect on the impact and flight of golf balls hit by a golf club. For this reason, the USGA regulates the surface roughness to ensure that the performance of a golf club is within a specified range such that a golf club cannot have such a high rate of performance that it would reduce the element of human skill in the sport of golf, and maintain the integrity of the sport and golf courses. The following is one example of USGA regulations related to roughness.

[0093] The roughness is determined at least in part by the diamond size, diamond density, and quality/uniformity of the coating. Different diamond sizes can influence roughness. With smaller diamond size, upon strike, the ball interfaces with the tips of the protruding diamond which is less surface area than a smoother/flatter club face. Kim was silent on what his roughness was, and was also silent on the density of the diamond in the coating. He claims the use of 1-6 micron diamond, but that is not enough to derive what roughness he would have had. Also, he did his work on flat panels while we used actual golf clubs that are entirely different than flat panels. The clubs were swinging, with grooves, and hitting the balls at an angle. This is very different than the dynamics of his dropping balls onto stationary flat plates. To create higher spin, we use larger diamond to create more. But using a coating with smaller diamond to preserve the face of a pre-roughened surface (like a wedge) would maintain the roughness as in example 9. Below. Kim never tested any golf clubs, and certainly not on any wedges where the objective is spin.

[0094] Different clubs are used for different purposes and, consequently, in the present disclosure different diamond sizes and density can be used for different clubs. In each case, however, uniformity is improved over Kim by way of use of PMSs. Different phosphorous baths also impact these club attributes.

[0095] 5.b (2)—Impact Area Roughness

[0096] When dealing with the surface roughness of a club face (not including putters, see Section 5f), the claims made by the manufacturer must be taken into account—especially if there is a claim that the roughness of the face influences the movement of the ball. In the absence of such claims, the ruling is made purely on the amount of roughness there is. Sandblasting or other treatments of roughness greater than 180 micro inches (4.5 μ metre) are not permitted. In addition to this requirement for roughness, milling is not permitted if the crest to trough depth exceeds 0.001 inches (0.025 mm). A reasonable tolerance is allowed for both of the above measurements. Non-conforming sandblasting or milling usually feels rough to the touch.

Corrosion Resistance of the Golf Club

[0097] Golf clubs must possess a significant degree of corrosion resistance. This is because golf clubs are subject to water and fertilizer materials on golf courses. Golf clubs are also known to be stored for long periods of times when not in use in areas that may be humid and prone to corrosion. In addition, golf clubs are routinely cleaned using water and other cleansers which is another reason why corrosion resistance is important. As was apparently unknown to Kim, composite coatings are known to be more prone to corrosion because of the presence of the particles in the metal alloy matrix, as such particles facilitate a pathway for corrosive elements to reach the base metal of the golf club and cause

corrosion. In the present disclosure, a higher phosphorus electroless nickel alloy potentially may be included to provide for more enhanced corrosion resistant properties. The present disclosure further includes the potential for applying an underlayer of a non-composite electroless nickel or other coating prior to the application of the composite electroless nickel coating. This underlayer can serve as a barrier layer against corrosion.

Longer Life of the Golf Club

[0098] The substantially improved composite coating of the present disclosure is capable of extending the life of the golf club to a greater extent than the concept expressed by Kim. Perhaps the greatest example of this improvement in the present disclosure is the vastly higher concentration or density of diamond capable in the present disclosure compared to Kim. Density of particles such as diamond play a key role in the wear resistance of the composite coating and therefore the lifetime of the golf club. In addition, Kim is silent on the uniformity of the diamond particles. In the composite coatings in the present disclosure, a substantially uniform distribution of diamond or other hard particles in the protein is achieved and therefore further leads to the longer life of the golf club, in addition to higher performance and more consistent performance. By hardening the composite coating by heat treatment after the coating is applied, as disclosed in the present disclosure, the coating will not only last longer, but can make the transfer of the impact energy of the golf club to the golf ball greater.

[0099] Conformity to rules established by agencies such as the United States Golf Association (USGA), Professional Golfers Association, and others that govern rules on equipment and other parameters of the sport.

[0100] This is a critical consideration necessary for composite coated golf clubs to be useful in the sport of golf. Professionals would not be permitted to use a golf club that does not conform to the rules of golf and potentially other regulations. Other agencies and organizations also specify that golf clubs must conform to USGA rules. This includes private golf clubs and tournaments. The composite coating of the present disclosure has been ruled by the USGA to conform to the rules of golf in USGA decision number 94-307. This ruling is included herein by reference. Even non-professional golfers would not want to play with equipment that does not conform to applicable rules, since doing so would make their performance and scores unworthy of comparison to other individuals and statistics where the rules were followed. Therefore, the market for non-conforming golf equipment would be so small that manufacturers of golf equipment would not generally want to produce non-conforming golf equipment, nor use a coating of any type that does not conform to applicable rules.

Environmental, Health and Safety Considerations in the Manufacture of Golf Clubs

[0101] These considerations are more of a concern now than at the time of Kim who was silent on such considerations. In the present disclosure, numerous aspects of the composite coatings demonstrated are far superior in terms of environmental, health and safety concerns. Coatings of the present disclosure can be made without the use of heavy metals such as lead or cadmium which were prominent in the plating industry at the time of Kim. Moreover, the present

disclosure can be made without the use of PFAS (per-and polyfluoroalkyl substances) materials which are of great current concern for environmental and health reasons as discussed herein. The present disclosure can also be made using a single component electroless nickel plating bath. This is advantageous because it is capable of allowing longer bath life, and therefore less waste. Such a single component electroless nickel plating bath can also be made and operated at less than six grams per liter of nickel metal that was dominant in the plating industry at the time of Kim. This is one more way to avoid environmental waste associated with the formation of the composite coating. Further, in the present disclosure, I disclosed that the density of the particles such as diamond in the composite protein can be as high as 50-60% by volume. This increase in diamond density not only improves the performance of the golf club, but it also reduces the amount of nickel and other materials used in the nickel phosphorus alloy of the coating.

Economy in the Manufacture of Golf Clubs

[0102] The use of the state-of-the-art plating solutions as in the present disclosure, in addition to the advantages noted herein, also serve to make the application of the composite coating on the golf clubs more economical and therefore commercially viable.

Improved bonding between the golf club head and the golf club shaft.

[0103] Golf club shafts are typically connected to the golf club heads by inserting the shaft into the hosel of the club head. The hosel is generally a cylindrical shaped tubular area of a golf club head into which the generally cylindrical shaft is inserted along with an epoxy type adhesive. The specific shape of the shaft, hosel and golf club may vary. A ferrule may also be used in the assembly of the shaft and club head. Proper adhesion of the shaft in the hosel is essential to the performance and durability of the golf club. Often when a shaft is inserted into a golf club head's hosel, along with the epoxy, some particulate matter may be incorporated into the epoxy. These particles may be glass beads, carbide, and other types of particles. They are often referred to as "shafting beads" in the golf industry. A purpose of these shafting beads is to fill some of the gap between the outside diameter of the shaft and the inside diameter of the hosel. The added surface area can also improve the adhesion of the shaft in the hosel. Kim is silent on this important aspect of golf club assembly and manufacture. The present disclosure has the benefit of improving the bonding of the shaft in the golf club head if the composite coating of the present disclosure is applied in the hosel. The coating provides multiple benefits including, but not limited to: 1) increased surface area on the inside diameter of the hosel on which the epoxy can bond with the shaft because of the textured surface of the coating of the present disclosure, 2) a better fit between the shaft and the hosel, and 3) the avoidance of any separate particulate matter such as shafting beads in the assembly of the shaft to the club head.

[0104] Though the present disclosure primarily focuses on golf clubs coated with composite electroless nickel phosphorus plating systems, there are other plating systems fall within the spirit of this disclosure. Other examples include, but are not limited to:

[0105] All electroless plating baths

[0106] All electroless nickel plating baths

[0107] Single or multiple component electroless nickel plating baths

[0108] EN systems using a replenishment and/or rejuvenation type process

[0109] All nickel-phosphorous alloy ratios

[0110] Electroless nickel-boron

[0111] Poly alloys

[0112] Electroless cobalt

[0113] EN systems with different levels of brightness

[0114] EN plating that is subsequently blackened

[0115] Non-metal stabilized plating systems

[0116] Metal stabilized plating systems

[0117] Heavy metal stabilized plating systems.

[0118] Coatings that conform to regulations and specifications such as REACH, RoHS, and the like.

[0119] Coatings with or without one or more polyfluoroalkyl substances (PFAS) materials.

[0120] Electroless copper, palladium, gold, and/or silver

[0121] Alloys/combinations thereof

[0122] All types of particulate matter that can be codeposited into electroless coatings.

[0123] The coatings on golf clubs of the present disclosure are also superior to, and serve additional unmet needs than, prior attempts to improve golf club performance with various materials.

[0124] For example, there was a commercialized golf club product using a carbide insert that created surface roughness. The most well-known company manufacturing wedges with a carbide insert was a company called Carbite Golf. While the objective of Carbite was the same as other products, there are multiple points as to why the present disclosure is a superior technology for golf clubs, and meets unmet needs. Carbite and the present disclosure are both methods to produce a defined surface roughness on the hitting surface. However, the present disclosure includes the possibility to use genuine diamond particles, the hardest material known to man. Carbite used silicon carbide particles that are not as hard and prone to breaking when impacted by a golf ball. These particles are impregnated into a soft bronze insert. The diamond particles of the present disclosure are incorporated in a much harder chemically applied nickel phosphorus alloy. This means that there are multiple layers of diamond uniformly dispersed within the coating. Therefore, as any diamond particles may get removed, more diamond below the surface will be revealed to ensure consistent roughness and performance.

[0125] The present disclosure is not DLC which means Diamond Like Carbon. The coatings of the present disclosure can contain genuine diamond, unlike DLC. The coatings of the present disclosure are also significantly different from DLC because these coatings can be made more economically, and with adjustable degrees of thickness, rigidity, hardness, flexibility, surface finish, and so on. Regarding rigidity, while the coatings of the present disclosure can be exceptionally hard, they can also have a high degree of flexibility. This is primarily due to the composite composition of the coatings of the present disclosure. The composite coatings can have more ductility than a metal alone. Therefore, golf clubs of the present disclosure can withstand the club head bending if the underlying club head is bent or otherwise distorted, such as in an operation to adjust the lie angle of the club head if needed by a golfer.

[0126] And we believe there is no issue of the club head bending if the underlying steel is cast or forged or even

milled. We also have successfully applied the coatings of the present disclosure to many different base metals.

[0127] And we believe there is no issue of the club head bending if the underlying steel is cast or forged or even milled. A previously produced, but now defunct, diamond-coated golf club wedge used in the 1990's was called PureSpin. PureSpin used industrial diamonds bonded to the clubface using an electrolytic nickel plating process. This is a significant difference than the coatings of the present disclosure which utilizes electroless plating. Electrolytic plating is not uniform in thickness like the coatings of the present disclosure, so it can distort the shape of the grooves and other surfaces. In the case of nickel coatings, electrolytic nickel plating is essentially pure nickel. The coatings of the present disclosure are alloys of nickel with other materials such as phosphorus. This allows the coatings of the present disclosure greater ability to have more properties that are useful on golf equipment, such as corrosion resistance, uniformity, flexibility, and other known differences between electroless nickel phosphorous, boron or other materials and electroplated nickel.

[0128] Liquidmetal golf clubs were made of a zirconium alloy that was promoted as significantly harder and stronger than titanium and steel and was promoted to have much higher energy transfer capabilities for golf clubs. The coatings of the present disclosure are significantly more economical than Liquidmetal. Moreover the coatings of the present disclosure can be produced in a range of parameters which can then meet a wider array of requirements on golf clubs.

[0129] With the present disclosure, it is possible to coat the entire golf club head or only select areas such as the clubface where impact is made with the golf ball, the bottom edge of the golf club head where impact and abrasion occurs with the grass, sand, debris, tees, and other surfaces on which a

robotic machine to simulate a genuine human swing of a golf club, but with more accuracy than a human. This testing was also performed on a field duplicating fairway conditions. All clubs (coated and uncoated) were constructed with Dynalite Gold® frequency matched shafts and had identical lofts, lies, head weights and swing weights. Precise center hits were done on all clubs with Titleist® DT90 balls. Weather conditions were ideal for testing.

[0131] This testing demonstrated an increase in ball distance of 7.34% from use of Composite Diamond Coating® golf club compared to an otherwise identical, but uncoated golf club. The increase in distance was seen in both the carry and total distance traveled by the golf balls. The standard deviation of the distances was also less with the Composite Diamond Coating® golf club. The Composite Diamond Coating® was 0.002 inches thick with a uniform distribution of diamond sized at 2 microns +/-0.2 microns at a density of 35-40% by volume in the coating. The surface of the composite coating on the hitting face was about 15 Ra. The coefficient of friction of the surface of the coating was about 0.3 The coating was produced in a medium phosphorous electroless nickel plating bath with the addition of the Composite Diamond Coating® CDC-D-2 diamond dispersion containing the 2 micron diamond particles in an aqueous dispersion with particulate matter stabilizers (PMSs), wherein the PMSs were free of PFAS materials. The golf club was heat treated at 350 degrees Celsius for two hours after coating to harden the matrix further, and optimize the adhesion of the coating to the base metal of the golf club.

[0132] The following illustrates one example from this testing. Additional tests were performed using different types of clubs (irons and woods) and different types of coatings including different sizes of diamond and different hard particles such as silicon carbide.

	CARRY DISTANCE	TOTAL DISTANCE	HEAD VELOCITY	BALL VELOCITY
UNCOATED DRIVER:				
Average	201.2 Yards	211 Yards	95.5 mph	194.38 Ft./Sec.
Standard Deviation	1.08 Yards	2.41 Yards	0.11 mph	0.37 Ft./Sec.
COMPOSITE DIAMOND COATED DRIVER:				
Average	207.6 Yards	226.5 Yards	95.6 mph	196.95 Ft./Sec.
Standard Deviation	0.8 Yards	2.16 Yards	0.05 mph	0.45 Ft./Sec.
Added Yards	6.4 Yards	15.5 Yards		
% Improvement	3.20%	7.34%		

golf club is swung. Any of these selections of areas to be coated can be made with the present disclosure in an economical manner.

EXAMPLES

Example 1

[0130] An independent laboratory was engaged to evaluate the performance of golf clubs featuring Composite Diamond Coating®, and versions of that coating were used in the examples of the present disclosure, where the coatings included PMSs, diamond particulate matter and, as appropriate, medium or high phosphorous, but without lead, cadmium, or other heavy metals, and no PFASs. This testing was performed outdoors on an electro-mechanical hitting

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Titleist is a registered trademark of Acushnet Company
Composite Diamond Coating is a registered trademark of Surface Technology, Inc.

One-Plate is a registered trademark of Surface Technology, Inc.

Example 2

[0133] The hitting face on the head of a wedge style golf club was coated with a Composite Diamond Coating® with a nickel-phosphorous alloy matrix that was 0.001 inches thick with a uniform distribution of diamond sized at 20 microns +/-2 microns at a density of about 50% by volume

in the coating. Prior to the coating, the stainless steel golf club was treated with a pretreatment process including alkaline cleaning, acid activation, an electrolytic nickel strike layer, and multiple rinsing steps. The surface of the composite coating on the hitting face had a coefficient of friction of about 0.5. The coating was produced in a One-Plate® 1001 medium phosphorous electroless nickel plating bath with the addition of the Composite Diamond Coating® CDC-D-20 diamond dispersion containing the 20 micron diamond particles in an aqueous dispersion with particulate matter stabilizers (PMSs), wherein the PMSs were free of PFAS materials. The golf club was heat treated at 350 degrees Celsius for two hours after coating to harden the matrix further, and optimize the adhesion of the coating to the base metal of the golf club.

[0134] This wedge head was then fitted onto a golf club shaft and the constructed golf club was then used to hit 300 golf balls. The spin rate was determined to be excellent for the intended purpose of a wedge. After these 300 golf ball hits, the surface of the hitting face of the golf club was examined under a microscope. No visible degradation of the surface could be seen between the area of the golf club face where the hits were made and the area of the club face where there were no impacts of golf balls. This demonstrates an exceptional wear resistance of the composite coating of the present disclosure. A photograph of this golf club is included herein as shown in FIG. 1.

Example 3

[0135] The head of a wedge style golf club was coated with two layers of electroless nickel coatings. The first layer applied to the golf club, after proper pretreatment of the base steel material, was a high phosphorous nickel-phosphorous alloy produced in a One-Plate® 2001 high phosphorous electroless nickel plating bath without the addition of any diamond or other particulate matter. After this initial layer, a second layer was applied on top of the first layer. The second layer was 0.002 inches thick with a uniform distribution of diamond sized at 40 microns $\pm$ 2 microns at a density of about 50% to 60% by volume in the coating. The second layer coating was produced in a One-Plate® 1001 medium phosphorous electroless nickel plating bath with the addition of the Composite Diamond Coating® CDC-D-40 diamond dispersion containing the 40 micron diamond particles in an aqueous dispersion with particulate matter stabilizers (PMSs), wherein the PMSs were free of PFAS materials. The golf club was heat treated at 350 degrees Celsius for two hours after coating to harden the matrix further, and optimize the adhesion of the coating to the base metal of the golf club. The surface of the composite coating on the hitting face had a coefficient of friction of about 0.7.

Example 4

[0136] The head of a wedge style golf club was coated with a coating that was 0.001 inches thick with a uniform distribution of diamond sized at 20 microns $\pm$ 2 microns at a density of about 50% to 60% by volume in the coating. All external surfaces of the golf clubhead were coated. The internal surface of the hosel was also coated. This coating was produced in a 650 gallon size plating bath of One-Plate® 1001 medium phosphorous electroless nickel plating bath with the addition of the Composite Diamond Coating® CDC-D-20 diamond dispersion containing the 20 micron

diamond particles in an aqueous dispersion with particulate matter stabilizers (PMSs), wherein the PMSs were free of PFAS materials, and such that the concentration of diamond particles in the plating bath was 7 grams per liter. The golf club was heat treated at 350 degrees Celsius for two hours after coating to harden the matrix further, and optimize the adhesion of the coating to the base metal of the golf club. The surface of the composite coating on the hitting face had a coefficient of friction of about 0.5 and a Ra roughness of about 145. A golf club shaft was then affixed to the golf club head by inserting the golf club shaft into the hosel of the golf club head with an epoxy type adhesive to bond the golf club shaft to the golf club head.

Example 5

[0137] The head of a wedge style golf club was coated with a coating that was 0.001 inches thick with a uniform distribution of diamond sized at 40 microns $\pm$ 2 microns at a density of about 50% to 60% by volume in the coating. All external surfaces of the golf clubhead were coated. The internal surface of the hosel was also coated. This coating was produced in a 140 gallon size plating bath of One-Plate® 1001 medium phosphorous electroless nickel plating bath with the addition of the Composite Diamond Coating® CDC-D-40 diamond dispersion containing the 40 micron diamond particles in an aqueous dispersion with particulate matter stabilizers (PMSs), wherein the PMSs were free of PFAS materials, and such that the concentration of diamond particles in the plating bath was 7 grams per liter. The golf club was heat treated at 350 degrees Celsius for two hours after coating to harden the matrix further, and optimize the adhesion of the coating to the base metal of the golf club. The surface of the composite coating on the hitting face had a coefficient of friction of about 0.7 and a Ra roughness of about 240. A golf club shaft was then affixed to the golf club head by inserting the golf club shaft into the hosel of the golf club head with an epoxy type adhesive to bond the golf club shaft to the golf club head.

Example 6

[0138] The head of a wedge style golf club was coated with a coating that was 0.001 inches thick with a uniform distribution of diamond sized at 20 microns $\pm$ 2 microns at a density of about 50% to 60% by volume in the coating. All external surfaces of the golf clubhead were coated. The internal surface of the hosel was also coated. This coating was produced in a One-Plate® 1001 medium phosphorous electroless nickel plating bath with the addition of the Composite Diamond Coating® CDC-D-20 diamond dispersion containing the 20 micron diamond particles in an aqueous dispersion with particulate matter stabilizers (PMSs), wherein the PMSs were free of PFAS materials.

[0139] A second layer was then applied over the above disclosed coating on the golf club head. This second layer was between 3-7 microns thick produced in a One-Plate® 1001 medium phosphorous electroless nickel plating bath.

[0140] The golf club was heat treated at 350 degrees Celsius for two hours after coating to harden the matrix further, and optimize the adhesion of the coating to the base metal of the golf club. The surface of the composite coating on the hitting face had a Ra roughness of about 136. A golf club shaft was then affixed to the golf club head by inserting

the golf club shaft into the hosel of the golf club head with an epoxy type adhesive to bond the golf club shaft to the golf club head.

Example 7

[0141] The head of a wedge style golf club was coated with a coating that was 0.001 inches thick with a uniform distribution of diamond sized at 40 microns $\pm$ 2 microns at a density of about 50% to 60% by volume in the coating. All external surfaces of the golf clubhead were coated. The internal surface of the hosel was also coated. This coating was produced in a One-Plate® 1001 medium phosphorous electroless nickel plating bath with the addition of the Composite Diamond Coating® CDC-D-40 diamond dispersion containing the 40 micron diamond particles in an aqueous dispersion with particulate matter stabilizers (PMSs), wherein the PMSs were free of PFAS materials.

[0142] A second layer was then applied over the above disclosed coating on the golf club head. This second layer was between 3-7 microns thick produced in a One-Plate® 1001 medium phosphorous electroless nickel plating bath.

[0143] The golf club was heat treated at 350 degrees Celsius for two hours after coating to harden the matrix further, and optimize the adhesion of the coating to the base metal of the golf club. The surface of the composite coating on the hitting face had a Ra roughness of about 196. A golf club shaft was then affixed to the golf club head by inserting the golf club shaft into the hosel of the golf club head with an epoxy type adhesive to bond the golf club shaft to the golf club head.

Example 8

[0144] The head of an iron style golf club was coated with a coating that was 0.002 inches thick with a uniform distribution of diamond sized at 2 microns $\pm$ 1 microns at a density of about 30 to 40% by volume in the coating, as well as a uniform distribution of a phosphorescent particulate material sized at about 2 to 10 microns at a density of about 2 to 5% by volume in the coating. All external surfaces of the golf clubhead were coated. The internal surface of the hosel was also coated. This coating was produced in a One-Plate® 1001 medium phosphorous electroless nickel plating bath with the addition of 1) the Composite Diamond Coating® CDC-D-20 diamond dispersion containing the 2 micron diamond particles in an aqueous dispersion with particulate matter stabilizers (PMSs), wherein the PMSs were free of PFAS materials, and 2) the TraceCoat-D dispersion containing the phosphorescent particles in an aqueous dispersion with particulate matter stabilizers (PMSs), wherein the PMSs were free of PFAS materials. The golf club was heat treated at 350 degrees Celsius for two hours after coating to harden the matrix further, and optimize the adhesion of the coating to the base metal of the golf club.

The surface of the composite coating on the hitting face had a coefficient of friction of about 0.5 and a Ra roughness of about 145. A golf club shaft was then affixed to the golf club head by inserting the golf club shaft into the hosel of the golf club head with an epoxy type adhesive to bond the golf club shaft to the golf club head.

[0145] The following photographs in FIG. 2 show the appearance of the golf club head of the Example 8 under 1) ultraviolet light (left), and 2) conventional LED ambient light (right). The red color resulting from the phosphorescent particles are clearly visible under ultraviolet light, yet not visible under conventional LED, fluorescent, incandescent, or sun light.

[0146] Example 9

[0147] Four identical TaylorMade “Milled Grind 4” 56-degree wedge style golf clubs were obtained. One of the four clubs was not altered from its original condition, as a control for the experiment of this example. The hitting face on three of the golf club heads were different versions of a Composite Diamond Coating® with a nickel-phosphorous alloy matrix that was 0.001 inches thick with co-deposited diamond sized at 2 microns $\pm$ 0.2 microns, 20 microns $\pm$ 2 microns, and 40 microns $\pm$ 2 microns, respectively. Prior to the coating, these three golf clubs were treated with a pretreatment process suitable for the base metal of the golf clubs. The coatings on these three golf clubs was produced in a One-Plate® 1001 medium phosphorous electroless nickel plating bath with the addition of the Composite Diamond Coating® CDC-D-2, CDC-D-20, and CDC-D-40 diamond dispersions, respectively. All such diamond dispersions contained particulate matter stabilizers (PMSs), wherein the PMSs were free of PFAS materials. The golf clubs were heat treated at 350 degrees Celsius for two hours after coating to harden the matrix further, and optimize the adhesion of the coating to the base metal of the golf club.

[0148] Each of the four golf clubs was then used to hit **600** golf balls. The spin rate was determined under controlled parameters by hitting identical Titleist Pro V1x golf balls in a Trackman golf testing equipment. Following this testing, the hitting faces of each of the four golf club heads was examined under a microscope. Significant wear to the initial texture/surface finish of the uncoated control golf club was observed. No noticeable wear was observed to the initial texture/surface finish of the hitting face of the three coated golf clubs. The effect on spin rate of balls hit by these four clubs is in the chart below. The data demonstrates that the uncoated control golf club lost 6.7% of its spin rate, the Composite Diamond Coated golf club with 2 micron diamond gained 0.3% more spin rate, the Composite Diamond Coated golf club with 20 micron diamond lost 0.4% of its spin rate, and the Composite Diamond Coated golf club with 40 micron diamond gained 1.8% more spin rate. See Tables 1 and 2 below.

TABLE 1

New- Uncoated Golf Club Face	Spin rate	New - CDC-2 Golf Club Face	Spin rate	New - CDC- 20 Golf Club Face	Spin rate	New - CDC- 40 Golf Club Face	Spin rate
1	8660	1	8857	1	8560	1	9500
2	9080	2	9991	2	9440	2	8752
3	9020	3	9419	3	8777	3	8942
4	9523	4	9326	4	8980	4	9233
5	8905	5	8539	5	9690	5	9028

TABLE 1-continued

New- Uncoated Golf Club Face	Spin rate	New - CDC-2 Golf Club Face	Spin rate	New - CDC- 20 Golf Club Face	Spin rate	New - CDC- 40 Golf Club Face	Spin rate
6	9173	6	8681	6	9410	6	9560
7	8964	7	9250	7	9120	7	9396
8	9393	8	9130	8	8990	8	9277
9	8860	9	8960	9	9347	9	8989
10	9253	10	9211	10	9120	10	8927
Average Spin Rate	9083		9136		9143		9160

TABLE 2

After 600 hits - Uncoated Golf Club Face	Spin rate	After 600 hits - CDC-2 Golf Club Face	Spin rate	After 600 hits - CDC-20 Golf Club Face	Spin rate	After 600 hits - CDC-40 Golf Club Face	Spin rate
1	8420	1	9140	1	9040	1	9450
2	8460	2	9130	2	9210	2	9260
3	8450	3	9358	3	9120	3	9458
4	8570	4	9010	4	9170	4	9327
5	8532	5	9100	5	9010	5	9510
6	8546	6	9260	6	9130	6	9170
7	8436	7	9345	7	9245	7	9320
8	8360	8	9180	8	9120	8	9234
9	8540	9	9056	9	9005	9	9310
10	8458	10	9056	10	9028	10	9236
Average	8477	Average	9164	Average	9108	Average	9328
Difference from new	-606		27		-36		167
Percentage change after 600	-6.7%		0.3%		-0.4%		1.8%

What is claimed is:

1. An improved golf club comprising at least a head and a shaft, the shaft being attached to the head, the head having a ball striking surface, wherein at least a central portion of the ball striking surface is covered with a nickel-based coating, and wherein said nickel-based coating further comprises diamond particulates with a uniform distribution, and said ball striking surface is of a Ra roughness of from 130 to 240.

2. The golf club of claim 1, wherein said coating is a medium phosphorous coating.

3. The golf club of claim 1, wherein said coating is a high phosphorous coating.

4. The golf club of claim 1, wherein said coating is applied in a controlled electroless nickel plating bath.

5. The golf club of claim 1, wherein said coating further includes particulate matter stabilizers.

6. The golf club of claim 1, wherein said nickel-based coating is absent of heavy metals and PFAS materials.

7. An improved golf club comprising at least a head and a shaft, the shaft being attached to the head, the head having a ball striking surface, wherein at least a central portion of the ball striking surface is covered with a nickel-based coating, and wherein said nickel-based coating further comprises diamond particulates with a uniform distribution.

8. The golf club of claim 7, wherein said coating is a medium phosphorous coating.

9. The golf club of claim 7, wherein said coating is a high phosphorous coating.

10. The golf club of claim 7, wherein said coating is applied in a controlled electroless nickel plating bath.

11. The golf club of claim 7, wherein said coating further includes particulate matter stabilizers.

12. The golf club of claim 7, wherein said nickel-based coating is absent of heavy metals and PFAS materials.

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